Python Interview Preparation

Hindi + English Answers | Fresher Data Analyst | Concepts, examples, practical interview responses

Complete practice set: 40 questions with English answer, Hindi explanation, and examples where useful.

How to use this PDF

First understand the Hindi explanation, then practice saying the English answer aloud. For code/formulas, explain what each part does. Adapt project claims to what you actually built and verified.

Q1. What is Python?

English answer: Python is a high-level, general-purpose programming language used for automation, data analysis, applications, and scripting.

Hindi explanation: Python easy-to-read programming language hai; automation, data analysis aur scripting me use hoti hai.

Q2. Why is Python used in data analytics?

English answer: It supports data cleaning, analysis, visualization, automation, and integration with databases and files through libraries.

Hindi explanation: Python se data clean, analyze, visualize aur repetitive tasks automate kar sakte hain.

Q3. What is a variable?

English answer: A variable is a name that refers to a value or object in memory.

Hindi explanation: Variable kisi value/object ko naam deta hai.

Example: `x = 10`

Q4. List vs tuple?

English answer: Lists are mutable; tuples are immutable sequences.

Hindi explanation: List change ho sakti hai; tuple create hone ke baad modify nahi hota.

Q5. What is a dictionary?

English answer: A dictionary stores key-value pairs and is useful for structured lookups.

Hindi explanation: Dictionary key-value pair me data rakhti hai.

Example: `person = {'name':'Asha','salary':4000}`

Q6. What is a set?

English answer: A set stores unique hashable values and is useful for removing duplicates or membership checks.

Hindi explanation: Set unique values rakhta hai; duplicates remove karne me useful hai.

Q7. What is the difference between == and is?

English answer: == compares values for equality; is checks whether two references point to the same object.

Hindi explanation: == value compare karta hai; is check karta hai ki dono same object hain ya nahi.

Q8. What are mutable and immutable types?

English answer: Lists and dictionaries are mutable; integers, strings, and tuples are immutable in their basic sense.

Hindi explanation: Mutable object modify ho sakta hai; immutable object ka value in-place change nahi hota.

Q9. What is a loop?

English answer: A loop repeats a block of code; for iterates over an iterable and while repeats while a condition is true.

Hindi explanation: Loop repeated task ke liye hota hai; for sequence par, while condition par chalta hai.

Q10. What is a function?

English answer: A function is a reusable block defined with def that can accept parameters and return a result.

Hindi explanation: Function reusable code block hai; parameters leta aur result return kar sakta hai.

Q11. What is a lambda?

English answer: A lambda is a small anonymous function, often used for short transformations.

Hindi explanation: Lambda short one-expression function hai.

Example: `lambda x: x*2`

Q12. What is list comprehension?

English answer: A concise way to create a list from an iterable, optionally applying a condition.

Hindi explanation: List comprehension se concise way me list create/filter karte hain.

Example: `squares = [x*x for x in range(5)]`

Q13. What is exception handling?

English answer: try/except handles expected runtime errors so a program can respond gracefully; finally can run cleanup.

Hindi explanation: try/except errors handle karta hai; finally cleanup ke liye ho sakta hai.

Q14. What is a module?

English answer: A module is a Python file containing reusable code that can be imported.

Hindi explanation: Module Python file hoti hai jisme reusable code hota hai.

Q15. What is a library?

English answer: A library is a collection of reusable modules and functions that provides capabilities such as analysis or plotting.

Hindi explanation: Library ready-made tools ka collection hai, jaise pandas data handling ke liye.

Q16. What is NumPy?

English answer: NumPy provides efficient arrays and numerical operations, especially for multidimensional data.

Hindi explanation: NumPy fast numerical arrays aur mathematical operations ke liye hai.

Q17. What is pandas?

English answer: pandas is a Python library for working with tabular data using Series and DataFrame.

Hindi explanation: Pandas rows-columns wale data ko DataFrame me clean/analyze karta hai.

Q18. Series vs DataFrame?

English answer: A Series is one-dimensional labeled data; a DataFrame is two-dimensional tabular data with rows and columns.

Hindi explanation: Series ek column jaisa; DataFrame rows aur columns wali table hai.

Q19. What is read_csv?

English answer: pandas.read_csv loads a CSV file into a DataFrame.

Hindi explanation: CSV file ko DataFrame me load karta hai.

Example: `df = pd.read_csv('employees.csv')`

Q20. loc vs iloc?

English answer: loc selects by labels and label-based conditions; iloc selects by integer position.

Hindi explanation: loc label/condition se select karta hai; iloc row-column ke number position se.

Example: `df.loc[df['salary']>3000, ['employee_name','salary']]`

Q21. How do you inspect a DataFrame?

English answer: Use head, tail, shape, info, describe, columns, and isna().sum() to understand rows, types, statistics, and missingness.

Hindi explanation: head(), shape, info(), describe() aur isna().sum() se data samjho.

Q22. How do you handle missing values?

English answer: First inspect why values are missing; then drop or impute using a justified rule and validate the impact.

Hindi explanation: Pehle missing ka reason samjho; phir justified rule se fill/drop karke impact check karo.

Q23. How do you remove duplicates?

English answer: Use duplicated() to inspect and drop_duplicates() to remove repeated rows, choosing subset and keep policy carefully.

Hindi explanation: duplicated() se identify, drop_duplicates() se remove; kaunsi columns par duplicate maana hai decide karo.

Q24. How do you filter salary above 3000?

English answer: Use a Boolean condition and select only required columns.

Hindi explanation: Condition lagao aur name/salary columns select karo.

Example: `df.loc[df['salary'] > 3000, ['employee_name','salary']]`

Q25. What is groupby?

English answer: groupby splits data into groups, applies aggregations, and combines results.

Hindi explanation: Department-wise average salary jaise summaries ke liye groupby use hota hai.

Example: `df.groupby('department')['salary'].mean()`

Q26. merge vs concat?

English answer: merge joins tables using keys; concat stacks objects along rows or columns.

Hindi explanation: merge common key se join; concat rows/columns me data jodta hai.

Q27. What is apply?

English answer: apply applies a function along a Series or DataFrame axis; vectorized operations are often faster for simple tasks.

Hindi explanation: apply custom function lagata hai; simple operations me vectorized method often faster hota hai.

Q28. What is a lambda in pandas?

English answer: A lambda can define a short transformation, often passed to apply or map.

Hindi explanation: Pandas me short custom transformation ke liye lambda use kar sakte hain.

Example: `df['name_length']=df['name'].apply(lambda x: len(str(x)))`

Q29. How do you sort data?

English answer: sort_values sorts by one or more columns; sort_index sorts by index.

Hindi explanation: Column ke basis par sort_values use hota hai.

Example: `df.sort_values('salary', ascending=False)`

Q30. What is Matplotlib?

English answer: Matplotlib is a plotting library for creating charts such as line, bar, scatter, and histogram plots.

Hindi explanation: Matplotlib Python me charts banane ki library hai.

Q31. What is Seaborn?

English answer: Seaborn builds statistical visualizations on top of Matplotlib with convenient defaults.

Hindi explanation: Seaborn statistical charts ko convenient syntax me banata hai.

Q32. How do you read Excel files?

English answer: Use pandas.read_excel, specifying sheet_name when needed; the engine dependency may be required.

Hindi explanation: Excel file pandas.read_excel se read karte hain; required engine installed hona chahiye.

Q33. How do you export results?

English answer: Use to_csv or to_excel, often with index=False when the DataFrame index is not a data column.

Hindi explanation: Result CSV/Excel me export kar sakte hain; unnecessary index avoid karne ke liye index=False.

Q34. What is a Jupyter Notebook?

English answer: An interactive environment that combines code, outputs, notes, and visualizations in cells.

Hindi explanation: Code, output, notes aur charts ko cells me run karne ka interactive environment hai.

Q35. What is an API?

English answer: An API is an interface that lets software systems exchange data or functionality; Python can call APIs using suitable libraries.

Hindi explanation: API systems ke beech data exchange ka interface hai; Python se API call kar sakte hain.

Q36. How do you connect Python to SQL?

English answer: Use a database driver or SQLAlchemy, parameterize queries, manage credentials securely, and close connections.

Hindi explanation: Database driver/SQLAlchemy se connect; credentials secure rakho aur parameterized query use karo.

Q37. What is a data-cleaning workflow?

English answer: Load data, inspect schema and quality, standardize types/labels, handle nulls/duplicates, validate, and export reproducibly.

Hindi explanation: Load → inspect → types/labels standardize → null/duplicate handle → validate → export.

Q38. How would you explain a Python project?

English answer: I used pandas to load and clean data, analyzed it with filtering and groupby, then exported validated outputs or visualized trends.

Hindi explanation: Pandas se data load/clean kiya, filter/groupby se analysis kiya aur output validate/export kiya.

Q39. What is the difference between shallow and deep copy?

English answer: A shallow copy creates a new outer container but may share nested objects; deep copy recursively copies nested objects.

Hindi explanation: Shallow copy me nested objects shared ho sakte hain; deep copy nested objects bhi copy karta hai.

Q40. What is PEP 8?

English answer: PEP 8 is the widely used Python style guide for readable and consistent code.

Hindi explanation: PEP 8 Python code ko readable aur consistent likhne ki style guide hai.

Interview tip: If you are unsure of a detail, explain your understanding honestly and describe how you would verify it.